Academic Psychiatry (2023) 47:676–679
https://doi.org/10.1007/s40596-023-01767-9

EDUCATIONAL CASE REPORT

The Role of Psychiatrists in Seclusion and Restraint Episodes: a Curriculum for Psychiatric Trainees

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Received: 11 November 2022 / Accepted: 3 March 2023 / Published online: 23 March 2023
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Psychiatrists have been involved in the use of seclusion and restraint for over 300 years [1]. Nearly as soon as the use of seclusion and restraint began in the 1700s, psychiatrists, such as Phillipe Pinel, were at the heart of reforming the practice [2]. The Consumer Movement in the 1960s and the key reports of abuse involving seclusion and restraint in the 1990s led to further emphasis on reform and oversight. However, the role of the psychiatrist in seclusion and restraint remains ill-defined and often varies based on particular institutional policies. Education for psychiatrists about the use of seclusion and restraint is also not standardized across institutions or training programs. As such, many psychiatrists may lack the necessary information and understanding to address the safe and patient-centered use of seclusion and restraint [3, 4].

The use of seclusion and restraint is intended to reduce imminent risk of harm to patients and staff. However, with each episode of seclusion and restraint, there is a risk of creating physical as well as psychological harm for patients. Restraints have been associated with a range of physical harm, including aspiration, asphyxiation, and cardiac complications. Psychologically, restraints have been linked to eliciting trauma responses, anxiety, nightmares, and distrust of mental health professionals [5]. This may be particularly true for patients who have a history of trauma [6]. Additionally, seclusion and restraint have been shown to be utilized disproportionately with Black patients when compared to their White counterparts [7]. The inequitable use of seclusion and restraint further exacerbates health disparities and compounds the impact of intergenerational racial trauma. Given the magnitude of these potential harms, the risks and benefits of each seclusion and restraint episode need to be considered carefully. Similar concerns apply to the use of chemical restraints, which are defined by Centers for Medicare and Medicaid Services as medication that is used to manage a patient's behavior or restrict freedom of movement, and is not a standard treatment or dosage for the patient's condition [8]. Chemical restraints are often utilized to manage patient agitation and confer similar risks to that of physical seclusion and restraint. However, they are not monitored or reported in the same manner as physical restraints [9].

As leaders of the multidisciplinary clinical team and prescribers of psychotropics, psychiatrists are uniquely positioned to weigh the risks and benefits related to physical and chemical restraints. Specific education dedicated to this topic is essential to integrate into training programs and continuing education to ensure psychiatrists engage in safe and effective practice. This may be particularly true for those working with children and adolescents as they are often restrained at higher rates and disproportionately experience harms [5, 10].

Here, the authors present a curriculum that was developed to improve knowledge and comfort for psychiatry trainees on an adolescent inpatient rotation in their role as psychiatrists managing episodes of seclusion and restraint. The curriculum includes a didactic component, interactive case discussion, and an open discussion around ethical considerations and racial disparities associated with seclusion and restraint. We present the curriculum and associated survey results examining changes in trainee knowledge and comfort.

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Curriculum and Survey Design

This class was designed for psychiatry interns. It begins with an approximately 60-min didactic section, utilizing PowerPoint slides as a guide. This segment provides a historical overview of the use of seclusion and restraint in the field of psychiatry [11]. Along with this historical context, it provides information on how various regulatory requirements came into existence for the use of physical restraints [8]. The concept of chemical restraint and the lack of current monitoring of this practice was also introduced [9, 12].

Utilizing this context, instructors guided learners through a 30-min open discussion around the ethical considerations involved in seclusion and restraint. This included a discussion about racial disparities and presentation of clinical and demographic data from the training institution about the use of physical restraints. The class ended with a group case discussion eliciting salient themes that often arise in seclusion and restraint episodes. There was an opportunity to discuss overarching tools utilized by various evidenced-based programs to reduce the use of seclusion and restraint, which provided residents with a framework to navigate these complex situations.

Prior to implementation, didactic content was reviewed by a fourth-year resident as well as two attending psychiatrists with expertise in the area. The class was presented by a fourth-year resident along with an attending psychiatrist from the inpatient adolescent unit. The learning objectives for the class are to review the role of the psychiatrist in seclusion and restraint through an historical lens, identify current regulations regarding indications, implementation and monitoring, recognize ethical factors, and identify opportunities for psychiatrists to play a role in reducing seclusion and restraint use. More detailed course materials are available from the corresponding author upon request.

Setting and Participants

This educational curriculum was designed to be a 90-min session that could be built into a weekly educational schedule for psychiatry interns during their 2-month exposure to adolescent psychiatry. Since this session became part of the existing educational structure, there was no additional advertisement or incentive to attend. This material was presented once during each 2-month block to a new set of psychiatry interns over a 2-year period. It was anticipated that approximately twenty-four learners would be exposed to this curriculum.

Survey Collection and Analysis

The authors utilized pre- and post-test assessments to determine the impact the curriculum had on trainees who experienced the class. The survey examined trainees' knowledge about the use of seclusion and restraint, comfort in being the psychiatrist involved with an episode of seclusion and restraint, and satisfaction with the utility of the educational content provided in the session.

Both the pre- and post-test surveys included the same six knowledge-based, multiple-choice questions. Both surveys also included a self-rated comfort measure ("How comfortable are you being the psychiatrist who orders the use of restraint or seclusion?"), rated on a Likert scale from 1 to 7 (extremely uncomfortable to extremely comfortable). Trainees were also asked if they felt they had received adequate education on the topic of seclusion and restraint, rated on a Likert scale from 1 to 5 (definitely inadequate to definitely adequate). Lastly, trainees were asked if they felt the information in this session was useful and would influence their future practice, both rated on a Likert scale from 1 to 5 (definitely not to definitely yes).

Learners were asked to complete the pre-test online prior to the start of the session and the post-test immediately following the session. Only those trainees who were present for the entire educational session completed both surveys.

Statistical analysis on the survey data was completed using Microsoft Excel. Paired *t*-tests were used to compare measures with pre- and post-test data and statistical significance was determined using a two-tailed alpha level of 0.05. This study was reviewed by the Yale University IRB and was deemed exempt from review.

Survey Results

A total of 24 learners were exposed to the didactic session between July 2020 and June 2022. Of these, 20 completed the pre-test, 18 completed the post-test, and 14 completed both the pre- and post-test, allowing for comparison of knowledge and comfort before and after the session. Results are summarized in Table 1 and described below.

On the pre-test ($N=20$), the majority reported receiving either a prior didactic session and/or informal teaching from an attending or resident on seclusion and restraint. Only 4 learners reported no prior education on this topic. However, on average, participants rated this prior amount of education to be inadequate for their level of training, at 2.3 out of 5 on a Likert scale.

The mean scores on the six knowledge-based questions significantly increased from 2.4 out of 6 (40%) on the pre-test to 5.2 out of 6 (87%) on the post-test ($N=14$, $p<0.01$). Self-rated comfort managing a seclusion and restraint event, measured on a Likert scale from 1 to 7, increased from 2.7 to 3.9 ($N=14$, $p=0.01$).

Regarding perceived usefulness of the didactic sessions, learners rated the session 4.7 out of 5 on the post-test ($N=18$). Finally, when asked whether the session would

Table 1 Pre- and post-test mean survey and assessment results

Question/questions	Pre-test	Post-test	*p*-value
Does the amount of prior education you have received on seclusion and restraint feel adequate for your level of training? (from 1 to 5)	2.3[a]		
How comfortable are you being the psychiatrist who orders the use of restraint or seclusion? (from 1 to 7)	2.7[b]	3.9[b]	0.01
Knowledge-based assessment (6 questions)	2.4 (40%)[b]	5.2 (87%)[b]	< 0.01
Was the didactic session on the use of seclusion and restraint useful in your learning? (from 1 to 5)		4.7[c]	
Will the content from this didactic session influence how you participate in seclusion and restraint events in the future? (from 1 to 5)		4.1[c]	

[a]Pre-test completion, $N = 20$

[b]Pre-test and post-test completion for comparison, $N = 14$

[c]Post-test completion, $N = 18$

influence their future practice, learners rated 4.1 out of 5 ($N = 18$).

Qualitative feedback was also solicited and was largely positive. The only consistent theme of constructive feedback that emerged was the desire for further case-based discussion to improve comfort with managing seclusion and restraint events in the future.

Reflection and Future Directions

This academic endeavor aimed to provide education and offer a space to facilitate dialogue on the use of seclusion and restraint in psychiatric practice. While this study includes a small number of learners, it serves as a proof of concept illustrating a curriculum in which learners showed advancement in their knowledge, significant improvements in their comfort managing these situations in the future, and a high-level of satisfaction with the session overall. The learners appreciated the ability to engage in safe dialogue about this practice, especially as it related to their current clinical cases.

Seclusion and restraint procedures sit at the interface between the freedom and rights of patients in the acute management of aggression and the maintenance of safety and security. Programs aimed at changing the culture of a psychiatric inpatient unit involve joining around a shared set of values, education and monitoring. The most successful effects on seclusion and restraint attitudes have been those that involve leadership, training, patient involvement, prevention tools, and a therapeutic environment [13]. This session involved trainees who may ultimately become psychiatric leaders and influence seclusion and restraint procedures in a variety of settings.

The facilitators of this workshop found that trainees were able to process the complexities around racial disparities and the intersection with re-traumatization by creating a safe space for case discussion. Participants requested more case-based discussion when asked for qualitative feedback and the facilitators often found this to be the richest component of the session. Though a prepared written case was presented in the didactic session, many trainees brought up cases of patients they were currently treating on clinical rotations who had been impacted by seclusion and restraint. In line with recently published Psychiatry Milestones from the Accreditation Council for Graduate Medical Education, trainees were engaged in an opportunity to reflect on their practice and understand the physician role in health care systems [14].

The authors acknowledge that this educational session has limitations. The results are from a small number of participants (24 total participants, with 14 completing both surveys) though this didactic series continues and collection of feedback is ongoing. Further, due to the hospital setting in which residents were working in at the time of these sessions, trainees often had acute clinical responsibilities which could impact their full participation. Thus, there was a number of individuals who were unable to complete both or either of the pre and post-surveys. Trainees who did complete both surveys did so immediately before and after this educational session. As such, growth in knowledge and comfort were measured near the time of learning. Future studies would benefit from testing knowledge and comfort levels after more time has passed. Additional limitations include that this workshop was only executed at one academic hospital center and in order to continue this series over academic years the session delivery and the style of engagement may have varied across individual presenters, though prior presenters trained future presenters to minimize variation.

Psychiatrists have always had a major role in determining when seclusion and restraint procedures are applied, reducing their use, and mitigating adverse outcomes to promote patient well-being. The quantitative feedback procured from this educational initiative speaks to the benefit of this training — results showed that residents were more comfortable in responding to situations of seclusion and restraint and

reported an increase in knowledge post lecture. Of note, even after the session, the level of comfort with seclusion and restraint procedures remained low. This educational module highlights the complexity of seclusion and restraint; thus, an improved but not overinflated level of comfort is promising. Further, results consistently showed that residents want more training and education about seclusion and restraint. Thus, future directions should consider the development of a whole educational series of several lectures and workshops — potentially incorporating seclusion and restraint prevention tools, evidence-based milieu management models, patient involvement, and leadership skills training. Response and observed enthusiasm suggest that interactive engagement around case discussions was received positively and thus future curriculum development should incorporate this feedback. Additionally, while significant changes in clinical outcomes were unlikely with this small-scale educational intervention, it may be beneficial in the future to collect clinical data on the frequency of seclusion and restraint episodes to determine whether sustained education of unit trainees has an impact on clinical outcomes.

Overall, survey results suggest that this workshop is successful in meeting its intended goals of improving the knowledge base and educational sophistication of future physician leaders in understanding the intricacies of seclusion and restraint. The key components of this workshop include promoting beneficial and patient-centered outcomes, starting with a historical perspective, foundational understanding of regulatory standards, interpersonal nuances of clinical relationships, and a safe space to invite challenging discussions around seclusion and restraint harms and disparities. Training institutions should ensure that they are adequately teaching future psychiatrists their role in minimizing the use and mitigating the harms of seclusion and restraint practices. The workshop described here is one example of an effective educational module.

Declarations

Disclosures On behalf of all authors, the corresponding author states that there is no conflict of interest.

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